

Machine Learning Based Integrated Solution for Smart Pepper Farming in Sri Lanka

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Abstract— Pepper (*Piper nigrum L.*) is one of Sri Lanka's major export crops. However, the industry faces several challenges such as yield uncertainty, inconsistent quality grading, price volatility and crop diseases. Most smallholder farmers still rely on traditional practices that lack accuracy and transparency. This research proposes an integrated intelligent system that utilizes Artificial Intelligence, Machine Learning and Blockchain Technologies to support data-driven decision making in pepper farming. The proposed solution consists of four main components that yield prediction using deep learning and environmental data, post-harvest quality grading using computer vision techniques, market price prediction with blockchain based supply chain transparency and pepper leaf disease detection using convolutional neural networks. These components are integrated into a mobile application named CeylonPepper, enabling farmers and exporters to access predictions, quality evaluation, disease diagnosis and verified market information through a single platform. Experimental results and preliminary field level validation indicate that the proposed models achieve stable and practical performance for real world applications. The integrated system aims to improve productivity, enhance quality assessment and increase transparency in the pepper value chain, contributing to more sustainable and technology-driven agriculture in Sri Lanka.

Keywords— Pepper farming, Machine Learning, Computer Vision, Blockchain, Smart Agriculture

I. INTRODUCTION

Pepper (*Piper nigrum L.*) is one of the most widely used and economically important spices in the world, valued for its distinctive flavor and aroma. In Sri Lanka, pepper is a major export crop that contributes significantly to the agricultural economy [1]. The country's tropical climate and fertile soil provide favorable conditions for cultivation. However, despite these advantages, productivity and quality remain inconsistent across different regions [2].

The pepper industry faces several challenges, including yield uncertainty, inconsistent quality grading, price volatility, limited supply chain transparency and difficulties in early

disease detection. These issues often lead to inefficient farm management, reduced product quality and lower market value for farmers. In addition, the lack of data-driven decision support systems makes it difficult for farmers and exporters to optimize production and marketing strategies. Existing solutions often address these problems individually, with limited integration into a single practical system.

This research proposes an integrated intelligent solution to address these challenges using machine learning and deep learning techniques. The system consists of four main components yield prediction using multimodal data and explainable AI techniques, post-harvest quality grading using computer vision, market price prediction combined with blockchain based traceability, and pepper leaf disease detection using deep learning models. These components are integrated into a mobile application named CeylonPepper, which serves as a unified platform for farmers and exporters.

Through this system, farmers can obtain yield predictions with interpretable insights, perform automated multi parameter quality assessment of pepper batches, access market price forecasts, and identify plant diseases at early stages. Preliminary evaluations with sample users indicate that the system can support practical decision making in real farming scenarios. By integrating these functionalities into a single platform, the proposed solution aims to support informed decision making, improve consistency in quality evaluation, and enhance transparency within the pepper value chain. Additionally, basic blockchain integration can be used to securely record transaction and quality data, ensuring data immutability, traceability, and trust among stakeholders in the supply chain.

II. LITERATURE REVIEW

Pepper farming in Sri Lanka faces several challenges including yield uncertainty, inconsistent quality grading, price volatility, and disease outbreaks. Traditionally, many of these decisions rely on manual observations and farmer experience, which are often subjective and lead to inaccurate

outcomes, particularly during early cultivation stages and post-harvest handling [3].

Early approaches to yield prediction mainly relied on statistical models such as regression and correlation analysis to study relationships between climatic factors and crop productivity [4]–[6]. While these methods provide moderate predictive performance, they are limited in capturing nonlinear relationships and temporal dependencies. More recent studies have applied machine learning models such as XGBoost and Random Forest along with explainable AI techniques like SHAP and LIME to improve both prediction accuracy and interpretability [7]–[9]. However, region-specific and practically deployable models for black pepper cultivation remain limited.

Post-harvest quality grading plays a key role in determining the commercial value of pepper in international markets. Standards such as ISO, Codex Alimentarius, and IPC define parameters including bulk density, extraneous matter, and contamination levels [10]–[12]. Despite these standards, grading is still largely manual, leading to inconsistent results. Previous studies have explored automated grading using image processing and machine learning techniques to detect impurities and classify quality based on visual features [13]–[15]. However, many of these approaches rely on handcrafted features or focus on limited quality parameters. Recent deep learning approaches, including object detection and lightweight convolutional neural networks, have improved robustness, though practical multi-parameter grading systems suitable for real farming environments are still limited.

Agricultural price predictions have also gained attention due to market instability affecting farmer income. Traditional time-series models such as ARIMA and SARIMA rely heavily on historical data and often fail to incorporate external variables such as weather conditions and economic indicators [16], [17]. Machine learning models, including ensemble methods and deep learning approaches like LSTM, have demonstrated improved performance by capturing complex patterns in data [18], [19]. In addition to prediction challenges, agricultural supply chains suffer from limited transparency and traceability. Blockchain technology has been proposed as a decentralized solution to ensure secure and tamper-resistant record-keeping, enabling stakeholders to verify product origin and transaction history [20], [21]. However, integrated solutions combining both price prediction and traceability in the pepper industry remain limited.

Plant disease detection is traditionally performed through expert-based visual inspection, which is time-consuming and not always accessible to farmers. Image processing techniques have been used to identify disease symptoms based on color and texture features [22]. More recently, deep learning models, particularly convolutional neural networks, have shown strong performance in automatically detecting and classifying plant diseases from leaf images [23]–[26]. Despite these advancements, research specifically focused on black pepper diseases and real-world deployment remains limited.

Overall, existing studies demonstrate the potential of machine learning, deep learning, and blockchain technologies in addressing individual agricultural challenges. However, most solutions focus on isolated problems such as yield prediction, quality grading, price forecasting, or disease detection, with limited attention to practical integration. There is a clear need for a unified system that combines these components into a single, scalable and user-friendly platform tailored to the Sri Lankan pepper industry. This research addresses this gap by proposing an integrated solution that supports end-to-end decision-making in pepper farming.

III. METHODOLOGY

A brief overview of the CeylonPepper approach is presented, followed by the methodology, outlining data handling, system development, and component integration. The multi-module architecture forms the foundation for a reliable solution for Sri Lankan pepper farming. Fig. 1 shows the overall system architecture, illustrating how the four main components are integrated within the mobile application. The integrated workflow enables all modules to operate within a single application, where output from each component contributes to the overall decision-support process. The system was designed with practical deployment considerations, allowing farmers to interact with the application in real usage scenarios.

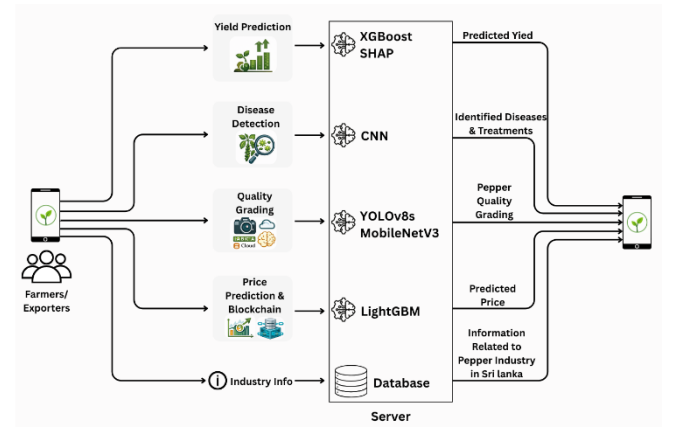


Fig. 1: System Overview

A. Yield Prediction

This study proposes a hybrid multimodal yield prediction framework that integrates deep convolutional feature extraction with structured environmental variables for black pepper yield estimation. The dataset consists of RGB images, soil moisture values, temperature measurements and corresponding yield data (kg) collected from three pepper plantations in the Kegalle District, Sri Lanka. All images were resized to 224×224 pixels and normalized using the EfficientNet preprocessing function. The dataset was partitioned into training (80%) and testing (20%) subsets using a fixed random seed to ensure reproducibility.

Transfer learning was employed using EfficientNetB0 pretrained on ImageNet, with the classification head removed (include_top=False) and global average pooling applied to

obtain fixed-length feature embeddings. The CNN was used solely as a feature extractor to mitigate overfitting due to limited data availability. In addition to raw environmental variables (soil moisture and temperature), nonlinear engineered features including moisture–temperature ratio, squared soil moisture and squared temperature terms were introduced to model potential nonlinear interactions.

To reduce redundancy in high-dimensional CNN features, Principal Component Analysis (PCA) was applied to retain 100 principal components, with fitting performed exclusively on the training set to prevent data leakage. Tabular features were standardized using StandardScaler prior to concatenation with PCA-reduced image features. The fused feature representation was used to train an XGBoost regressor, selected for its strong predictive performance, built-in regularization and capability to model nonlinear feature interactions. To evaluate the impact of feature engineering and dimensionality reduction, a baseline XGBoost model using raw features without PCA or derived variables was implemented for comparison.

Model performance was assessed using the coefficient of determination (R^2), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Model interpretability was enhanced using SHAP to quantify feature contributions. Given that the dataset size is limited, regularization techniques and feature engineering were applied to improve generalization performance.

B. Post-Harvest Pepper Quality Grading

The proposed post-harvest pepper quality grading module integrates computer vision techniques along with user-provided physical measurements to evaluate multiple quality parameters of black pepper batches. The objective is to provide a consistent and automated grading mechanism that considers both physical and visual quality characteristics in accordance with international standards.

The grading process begins with batch information collection through the mobile application, where farmers provide details such as pepper variety, harvest date, drying method and batch weight. These details are stored in the system database and associated with the corresponding batch record. Bulk density is measured by farmers using a standard 1 L container and the recorded value is entered into the application for further analysis.

For visual analysis, the system requires nine images captured from three locations within the pepper sack: top, middle and bottom. Each location includes full view, half view and close-up images to ensure representative sampling and reduce bias. The images are processed using a YOLOv8s object detection model, which identifies pepper berries, adulterant seeds and extraneous matter. Detected pepper berries are cropped and passed to two MobileNetV3-based classifiers to detect mold contamination and abnormal textures such as damaged or broken berries. MobileNetV3 was selected due to its efficiency and suitability for real-time applications.

Based on detection and classification results, quality metrics including adulterant seed percentage, extraneous

matter percentage, mold contamination, abnormal texture and healthy berry percentage are calculated. Adulterant and extraneous matter percentages are computed relative to total detected objects, while other metrics are calculated relative to the total number of pepper berries.

Results from three images per location are averaged to obtain sample-level values, and final batch metrics are derived by averaging the top, middle and bottom samples. These values are then processed by a grading engine that assigns weighted scores based on predefined thresholds. The final score determines the quality grade and generates a detailed report with improvement recommendations for farmers.

C. Market Price Prediction and Blockchain Integrity

The proposed system integrates machine learning-based price prediction with blockchain-based supply chain verification to support decision-making and transparency in the Sri Lankan pepper industry. The methodology focuses on two main components: market price prediction and blockchain-based traceability of pepper batches.

Historical data was collected from multiple sources. Local market price data were obtained from the Department of Agriculture, while export price data and economic indicators such as export volume and exchange rates were collected from the Central Bank of Sri Lanka. Weather data including temperature, humidity, rainfall and wind speed were obtained from the Open-Meteo API. These datasets were combined to form a multi-variable dataset.

Before training, preprocessing steps were applied including handling missing values and encoding categorical variables such as district, pepper type and grade. Feature engineering was used to extract relevant attributes. For local price prediction, inputs included environmental factors and time-based variables, while export price prediction used economic indicators and temporal features.

Several machine learning models, including Random Forest, XGBoost, CatBoost and LightGBM were evaluated. The selected model forecasts weekly local prices and monthly export prices. Based on predictions, the system provides decision support such as whether to sell or wait.

To ensure transparency, each pepper batch created by the farmer is recorded as a secure data block within the system. Farmers can also list their batches on the marketplace; however, both batch creation and marketplace listing are subject to admin verification. Once the details are validated and approved, the admin generates a unique QR code for each pepper batch.

Exporters can view comprehensive details related to each approved pepper batch through the system interface, including origin, quality assessment data, and price information from the date of harvest onward. A QR code is also displayed within the batch details to support future traceability features. This approach enhances transparency and trust in the pepper supply chain by ensuring that all verified information is consistently accessible and prepared for future end-to-end traceability.

D. Pepper Leaf Disease Detection

This research focuses on developing a deep learning-based system for detecting black pepper diseases and providing digital support for farmers. The system is designed to identify common diseases while enabling communication and knowledge sharing.

A dataset of black pepper leaf images was collected covering five diseases: Yellow Mottle, Leaf Blight, Pollu Disease, Slow Decline and Foot Rot. Images were obtained under varying environmental conditions to improve robustness. Preprocessing included resizing, normalization and noise reduction. Data augmentation techniques such as rotation, flipping and zooming were applied to increase dataset diversity and reduce overfitting.

A Convolutional Neural Network (CNN) was used to classify images into disease categories. The model was trained over multiple epochs to learn distinguishing visual features. Evaluation was performed using a separate test dataset with metrics including accuracy, precision, recall and F1-score.

In addition to disease detection, the system includes a digital platform with knowledge sharing and complaint management features. Farmers can share experiences and report issues, which are reviewed by agricultural officers through an administrative interface.

All components are integrated into a single platform, supporting disease detection, information access and communication for improved decision-making.

E. Mobile Application Interface

The developed CeylonPepper mobile application provides a user-friendly interface for farmers and exporters to interact with the system. Fig. 2 shows selected screens of the application, including the dashboard, yield prediction, quality grading and disease detection interfaces.

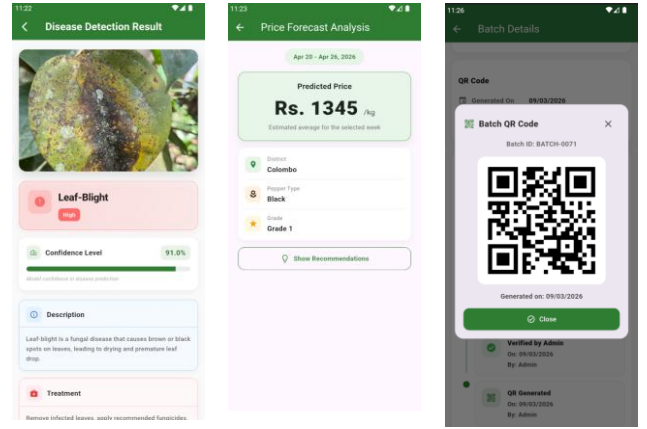
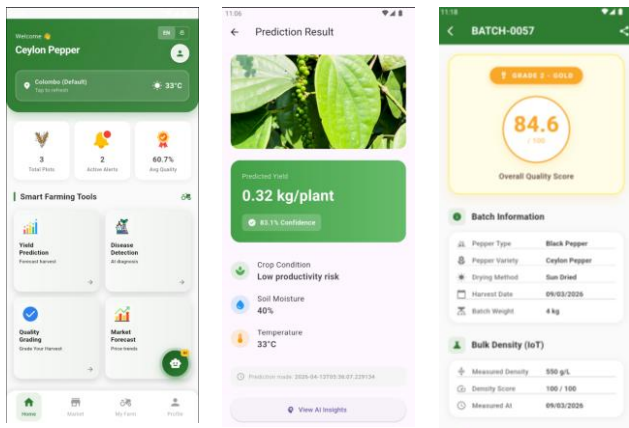


Fig. 2: CeylonPepper Mobile Application Screens

IV. RESULTS AND DISCUSSION

To provide a clear comparison of all components, Table 1 summarizes the key performance metrics of the proposed system modules.

TABLE I: Summary of Model Performance Across Components

Component	Model	Key Metrics
Yield Prediction	EfficientNetB0 + XGBoost	Train R ² : 0.88, Test R ² : 0.85, RMSE: 0.42, MAE: 0.31
Quality Grading	YOLOv8s (Detection)	mAP50: 0.837, mAP50-95: 0.595
	MobileNetV3 (Classification)	Mold Acc: 89.2%, Texture Acc: 87.6%
Price Prediction	LightGBM	Accuracy: 87.97%, MAPE: 12.02%, R ² = 0.87
Disease Detection	CNN	Accuracy: 88.5%, F1-score: 0.87

These results indicate that all modules achieve stable and practical performance levels suitable for real-world agricultural applications. The balanced performance across components demonstrates the effectiveness of the integrated system rather than relying on a single model.

A. Yield Prediction

To evaluate the proposed yield prediction system, experiments were conducted using a dataset consisting of early-stage pepper plant images and environmental variables, including soil moisture and temperature. Visual features extracted using EfficientNetB0 were combined with environmental data, and PCA was applied to reduce feature dimensionality. The final model was implemented using XGBoost. The dataset was divided into training and testing subsets using an 80:20 split, and performance was evaluated using MAE, RMSE and R².

The model achieved an R² score of 0.88 on the training set and 0.85 on the testing set, indicating good generalization with a minimal performance gap. The RMSE and MAE values further confirm that the model can reasonably predict yield values under varying conditions.

Explainable AI techniques using SHAP were applied to interpret predictions. The analysis showed that soil moisture is the most influential factor, followed by temperature and image-based features related to plant health. These findings are consistent with domain knowledge and improve model transparency.

Validation with a small group of farmers indicated that the predicted yield ranges were aligned with expected harvest outcomes, demonstrating the practical applicability of the model. However, performance is still influenced by dataset size, and future work should include additional environmental variables such as rainfall and humidity.

B. Post-Harvest Pepper Quality Grading

The quality grading module was evaluated using images collected from real pepper samples under controlled conditions with a grid-based white background. The dataset consisted of a large number of labeled objects, with class distribution as follows: pepper berries (3640), adulterant seeds (939) and extraneous matter (253). To address class imbalance, oversampling techniques were applied to minority classes.

The YOLOv8s model achieved an mAP50 of 0.837 and mAP50–95 of 0.595, indicating reliable detection performance. Pepper berry detection achieved high precision (0.967) and recall (0.949), while other classes also showed satisfactory detection capability.

For defect classification, MobileNetV3 models were trained on cropped berry images. The mold detection model achieved an accuracy of 89.2%, while the texture classification model achieved 87.6%, demonstrating good performance in identifying defective berries without overfitting.

During inference, the system analyzes nine images per batch (top, middle and bottom). The computed quality metrics include adulterant percentage, extraneous matter percentage, mold contamination, abnormal texture and healthy berry ratio. A sample batch evaluation produced values of 7.82% adulterant seeds, 7.70% extraneous matter, 14.19% mold, 31.13% abnormal texture and 54.67% healthy berries.

These values are used by the grading engine to assign a final quality score and grade based on predefined standards. Validation with users showed that the automated grading results are consistent with manual inspection trends while providing more objective and repeatable measurements.

C. Market Price Prediction and Blockchain Integrity

Price Prediction was evaluated using historical market, environmental and economic data. Several machine learning models, including Random Forest, XGBoost, LightGBM and CatBoost, were tested, and LightGBM was selected as the final model due to its superior performance. The model achieved an accuracy of 87.97% with a MAPE of 12.02%,

indicating strong predictive capability with relatively low error.

The performance comparison of all models based on Accuracy and MAPE is presented in Fig. 3, which provides a clear visual representation of model effectiveness and highlights the differences in prediction accuracy and error rates.

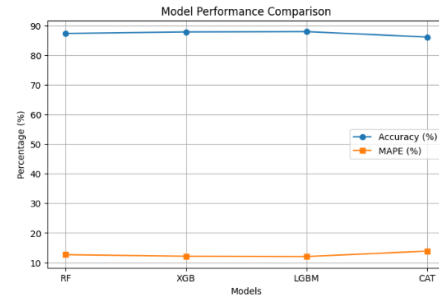


Fig. 3: Comparison of machine learning models for pepper price prediction

Feature importance analysis revealed that export prices, exchange rates, seasonal patterns and weather variables significantly influence price fluctuations. The system provides weekly local price forecasts and monthly export price predictions, along with simple decision recommendations for users.

The blockchain component was evaluated through functional testing by recording and retrieving batch data using QR codes. The system ensured data immutability and traceability, allowing users to verify batch origin and transaction history. User-level testing indicated that the blockchain integration improves transparency while remaining easy to use in practical scenarios.

D. Pepper Leaf Disease Detection

The disease detection model was evaluated using a dataset of black pepper leaf images representing five disease classes. The CNN model was trained and tested using standard classification procedures.

The model achieved an overall accuracy of 88.5%, with balanced performance across precision, recall and F1-score (approximately 0.87). These results indicate that the model can reliably distinguish between different disease categories under varying conditions.

Training and validation accuracy improved steadily without significant overfitting. The model successfully captured visual features such as discoloration, texture variation and leaf damage patterns.

Initial validation with users showed that the system can correctly identify common diseases in most cases, providing useful support for early diagnosis. However, performance can be further improved by increasing dataset size and including more diverse field conditions.

V. CONCLUSION

This research presents an integrated intelligent system to support decision-making in pepper farming in Sri Lanka. The proposed solution combines machine learning and deep learning technologies to address key challenges including yield prediction, quality assessment, price forecasting and disease detection within a single mobile platform.

The system provides reliable yield estimation, automated multi-parameter quality grading, data-driven price forecasting and early disease detection, while improving transparency through blockchain based traceability.

Experimental results demonstrate that all modules achieve stable and practical performance under real-world conditions, with initial user level validation indicating consistency with actual farming practices.

Overall, the proposed system highlights the potential of AI driven technologies to improve productivity, consistency and transparency in the pepper value chain. Future work will focus on expanding datasets, improving model robustness and conducting large-scale field validation.

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